

TOPOLOGICAL CYCLIC HOMOLOGY OF RINGS OF INTEGERS IN p -ADIC FIELDS VIA SYNTOMIC COHOMOLOGY

WEI YANG

ABSTRACT. Let $K/\mathbf{Q}_p$ be a finite extension, with valuation ring $\mathcal{O}_K$. Taking the calculation of the BMS syntomic complexes $\mathbf{Z}_p(n)(\mathcal{O}_K)$ from the companion note [8] as arithmetic input, we compute

$$\pi_*(\mathrm{TC}(\mathcal{O}_K; \mathbf{Z}_p) \wedge \mathbb{S}/p)$$

from the Bhatt–Morrow–Scholze motivic filtration. We prove degreewise strong convergence, identify the two-step filtration in every positive even stem with the Moore coefficient sequence, and resolve all hidden additive extensions. For odd p the Moore spectrum has exponent p , so the extensions split. At $p = 2$, the relation $2 \mathrm{id}_{\mathbb{S}/2} = i\eta q$ reduces the problem to the integral product of the Hopf class with the unique two-torsion class in an odd stem. The image of η is $-1 \in \widehat{\mathcal{O}_{K,2}^\times}$, and the obstruction is the quaternion symbol $\{-1, -1\}_K$. Thus $\pi_{2m}\mathrm{TC}(\mathcal{O}_K; \mathbf{F}_2) \cong \mathbf{Z}/4$ exactly when both m and $[K : \mathbf{Q}_2]$ are odd. No relative-THH descent calculation is used.

1. INTRODUCTION

Let p be a prime and let $K/\mathbf{Q}_p$ be a finite extension. Write

$$e = e_K, \quad f = f_K, \quad N = [K : \mathbf{Q}_p] = ef,$$

and let k be the residue field of K . Write $\chi : G_K \rightarrow \mathbf{Z}_p^\times$ for the p -adic cyclotomic character. We use the notation

$$X_K := \mathrm{TC}(\mathcal{O}_K; \mathbf{Z}_p),$$

where p -completion is understood as Bousfield localization with respect to the Moore spectrum $\mathbb{S}/p$; for the bounded-below spectra occurring here this agrees with the usual derived p -completion. For any spectrum Y and $r \geq 1$, write

$$Y/p^r := Y \wedge \mathbb{S}/p^r.$$

Thus $\mathrm{TC}(\mathcal{O}_K; \mathbf{F}_p)$ means the Moore-spectrum reduction X_K/p ; it does not mean tensor product with the Eilenberg–MacLane spectrum $H\mathbf{F}_p$. For an

Date: July 2026.

2020 *Mathematics Subject Classification.* Primary 19D55; Secondary 11S25, 14F30, 55T25.

Key words and phrases. topological cyclic homology, motivic filtration, syntomic cohomology, local fields, Hilbert symbol.

abelian group A , the symbols A/p^r and $A[p^r]$ mean $A/(p^r A)$ and $\ker(p^r : A \rightarrow A)$, respectively. Parentheses will always be inserted when a homotopy group, rather than a spectrum, is being reduced modulo p^r . Complexes are cohomologically indexed, and all tensor products of complexes are derived unless explicitly indicated otherwise.

Main results. For $a \in \mathbf{Z}$, set

$$\varepsilon_a := \dim_{\mathbf{F}_p} H^0(K, \mathbf{F}_p(a)).$$

If p is odd and $d = [K(\mu_p) : K]$, then

$$\varepsilon_a = \begin{cases} 1, & d \mid a, \\ 0, & d \nmid a. \end{cases} \quad (1)$$

If $p = 2$, then $\varepsilon_a = 1$ for every a , since $\mathbf{F}_2^\times$ is trivial.

Theorem 1.1 (Mod- p calculation). *The groups $\pi_t(X_K/p)$ vanish for $t < -1$, and*

$$\pi_{-1}(X_K/p) \cong \mathbf{F}_p, \quad \pi_0(X_K/p) \cong \mathbf{F}_p.$$

For odd stems,

$$\begin{aligned} \pi_1(X_K/p) &\cong \mathbf{F}_p^{N+\varepsilon_1}, \\ \pi_{2m-1}(X_K/p) &\cong \mathbf{F}_p^{N+\varepsilon_m+\varepsilon_{1-m}} \quad (m \geq 2). \end{aligned}$$

For $m \geq 1$, the positive even stems are as follows.

(i) If p is odd, then

$$\pi_{2m}(X_K/p) \cong \begin{cases} \mathbf{F}_p \oplus \mathbf{F}_p, & d \mid m, \\ 0, & d \nmid m. \end{cases}$$

(ii) If $p = 2$, then

$$\pi_{2m}(X_K/2) \cong \begin{cases} \mathbf{Z}/4, & m \text{ odd and } [K : \mathbf{Q}_2] \text{ odd}, \\ \mathbf{F}_2 \oplus \mathbf{F}_2, & \text{otherwise.} \end{cases}$$

In particular,

$$\pi_1(X_K/2) \cong \mathbf{F}_2^{N+1}, \quad \pi_{2m-1}(X_K/2) \cong \mathbf{F}_2^{N+2} \quad (m \geq 2).$$

The same argument computes integral p -complete topological cyclic homology. For $a \neq 0$, put

$$W_a(K) := H^0(K, \mathbf{Q}_p/\mathbf{Z}_p(a)), \quad A^\vee := \text{Hom}_{\text{cont}}(A, \mathbf{Q}_p/\mathbf{Z}_p).$$

The group $W_a(K)$ is a finite cyclic p -group: choose $g \in G_K$ with $\chi(g)^a \neq 1$; then every invariant in $\mathbf{Q}_p/\mathbf{Z}_p(a)$ is annihilated by the nonzero p -adic integer $\chi(g)^a - 1$, and every finite subgroup of $\mathbf{Q}_p/\mathbf{Z}_p$ is cyclic. Moreover,

$$W_a(K)[p] = H^0(K, \mathbf{F}_p(a)),$$

so $W_a(K)$ is nonzero exactly when $\varepsilon_a = 1$.

Theorem 1.2 (Integral calculation). *One has $\pi_t X_K = 0$ for $t < -1$ and*

$$\pi_{-1} X_K \cong \mathbf{Z}_p, \quad \pi_0 X_K \cong \mathbf{Z}_p, \quad \pi_1 X_K \cong \widehat{\mathcal{O}_{K,p}^\times}.$$

For $m \geq 2$ there is a canonical identification

$$\pi_{2m-1} X_K \cong H^1(K, \mathbf{Z}_p(m)),$$

and, noncanonically,

$$H^1(K, \mathbf{Z}_p(m)) \cong \mathbf{Z}_p^N \oplus W_m(K).$$

For every $m \geq 1$,

$$\pi_{2m} X_K \cong W_{-m}(K)^\vee.$$

Strategy and independence from the relative-THH calculation. The proof deliberately separates the arithmetic input from the filtered-homotopy argument. The arithmetic input is the calculation of the complexes

$$C_n := \mathbf{Z}_p(n)(\mathcal{O}_K)$$

from the companion note [8]. We recall only its precise statement, together with the multiplicative compatibility needed later. In particular, we do not repeat the comparison with Sato twists, the localization calculation on the trait, or the local Galois-cohomology computation.

The purpose of the present paper is to make the passage from these complexes to the abutment completely explicit. We first construct and index the integral and Moore-reduced motivic spectral sequences and prove strong convergence degree by degree. We then identify, canonically, the two-step filtration in every positive even mod- p stem with the short exact sequence obtained from the Moore cofiber sequence. This identification is essential at $p = 2$: it makes the hidden extension a concrete stable-homotopy operation, rather than an unspecified extension of two $\mathbf{F}_2$ -vector spaces. The remaining calculation is the formula

$$2x = i_*(\eta \partial x)$$

for a Moore lift x , followed by a filtered-product computation of $\eta \partial x$.

The formulas agree with Liu–Wang’s computation [6, Theorems 8.20 and 8.21]. Their paper is cited only for comparison: no result from their relative-THH descent spectral sequence is used here. This independence claim is deliberately limited: the proof still relies on the BMS motivic filtration, the Bhatt–Mathew comparison, the companion syntomic calculation, the Clausen–Mathew–Morrow finite-coefficient comparison, and standard local Galois cohomology.

The logical dependence on the companion note is exact: the present paper uses Theorem 1.2, Corollary 1.3, and Proposition 1.6 of the June 2026 version of [8], together with the finite-level multiplicativity supplied by [2, Theorem 5.8]. In particular, the present calculation should be read together with that companion preprint until its arithmetic input has appeared in a stable archival form.

Section 2 treats the filtered spectrum and convergence. Section 3 records the syntomic input from [8] and derives the finite-coefficient page. Section 4 computes the E_∞ -page and identifies the canonical extension sequence in each stem. Section 5 resolves the odd-primary extensions. Section 6 gives the detailed 2-primary calculation, including the image of η , the cyclotomic Bockstein, compatibility of filtered multiplication with Galois cup products, and the Hilbert-symbol test.

2. THE MOTIVIC FILTRATION AND ITS INDEXING

We recall the topological input. Throughout, $\mathbf{Z}_p(n)(A)$ denotes the BMS syntomic complex in weight n . The ring $\mathcal{O}_K$ is quasisyntomic. Indeed, if $K_0/\mathbf{Q}_p$ is the maximal unramified subextension of $K/\mathbf{Q}_p$, then $\mathcal{O}_{K_0}$ is finite étale over $\mathbf{Z}_p$ and $\mathcal{O}_K \cong \mathcal{O}_{K_0}[T]/(E(T))$ for an Eisenstein polynomial E ; hence $\mathcal{O}_K$ is a p -complete local complete-intersection $\mathbf{Z}_p$ -algebra.

Lemma 2.1 (Finite Moore coefficients and p -completion). *For every spectrum Y and every $r \geq 1$, the p -completion map induces an equivalence*

$$Y/p^r \xrightarrow{\sim} Y_p^\wedge/p^r.$$

In particular,

$$\mathrm{TC}(\mathcal{O}_K)/p^r \xrightarrow{\sim} X_K/p^r.$$

Proof. The cofiber of the localization map $Y \rightarrow Y_p^\wedge = L_{\mathbb{S}/p}Y$ is $\mathbb{S}/p$ -acyclic. The finite Moore spectrum $\mathbb{S}/p^r$ belongs to the thick tensor ideal generated by $\mathbb{S}/p$: this follows by induction from the octahedral cofiber sequences relating $\mathbb{S}/p^{r-1}$, $\mathbb{S}/p^r$, and $\mathbb{S}/p$. Hence the cofiber becomes contractible after smashing with $\mathbb{S}/p^r$, which is precisely the asserted equivalence. $\square$

Theorem 2.2 (Bhatt–Morrow–Scholze). *Let A be a quasisyntomic ring. The p -complete spectrum $\mathrm{TC}(A; \mathbf{Z}_p)$ admits a complete, exhaustive, decreasing, multiplicative filtration $\mathrm{Fil}_{\mathrm{mot}}^\bullet$ such that*

$$\mathrm{gr}_{\mathrm{mot}}^n \mathrm{TC}(A; \mathbf{Z}_p) \simeq \mathbf{Z}_p(n)(A)[2n].$$

The complexes $\mathbf{Z}_p(n)(A)$ vanish for $n < 0$, and the associated spectral sequence is

$$E_2^{i,j} = H^{i-j}(\mathbf{Z}_p(-j)(A)) \implies \pi_{-i-j} \mathrm{TC}(A; \mathbf{Z}_p). \quad (2)$$

Proof. The complete, exhaustive, decreasing, multiplicative filtrations on THH , TC^- , and TP are constructed in [3, Theorem 1.12(2)]; the filtered fiber construction for TC and the formula for its graded pieces are [3, Theorem 1.12(5)]. More precisely, the canonical and Frobenius maps are maps of filtered E_∞ -rings, so their equalizer in filtered E_∞ -rings supplies the asserted multiplicative filtration on TC ; on underlying filtered spectra this equalizer is the displayed homotopy fiber of their difference. The filtration is $\mathbf{Z}$ -indexed and exhaustive in the negative direction. We use the BMS convention $\mathbf{Z}_p(n) = 0$ for $n < 0$, also adopted in [8, Section 1]. $\square$

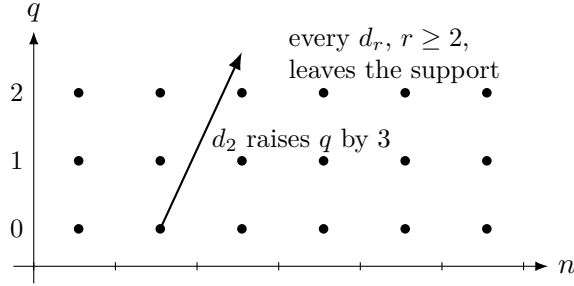


FIGURE 1. The formal degeneration mechanism. Only the cohomological rows $q = 0, 1, 2$ occur.

Put

$$C_n := \mathbf{Z}_p(n)(\mathcal{O}_K), \quad \overline{C}_n := C_n \otimes_{\mathbf{Z}_p}^{\mathbf{L}} \mathbf{F}_p.$$

Define the filtration on the Moore reduction levelwise by

$$\mathrm{Fil}_{\mathrm{mot}}^n(X_K/p) := \mathrm{Fil}_{\mathrm{mot}}^n X_K \wedge \mathbb{S}/p.$$

Since $\mathbb{S}/p$ is finite and therefore dualizable, smashing with it commutes with the inverse limits entering completeness; it also commutes with all colimits and cofibers. Consequently this filtration is complete and exhaustive, and

$$\mathrm{gr}^n(X_K/p) \simeq \overline{C}_n[2n].$$

Proposition 2.3 (Reindexing). *The integral and mod- p motivic spectral sequences can be written as*

$$\mathcal{E}_2^{n,q} = H^q(C_n) \implies \pi_{2n-q} X_K, \quad (3)$$

$$\overline{\mathcal{E}}_2^{n,q} = H^q(\overline{C}_n) \implies \pi_{2n-q}(X_K/p). \quad (4)$$

In these coordinates,

$$d_r : \mathcal{E}_r^{n,q} \longrightarrow \mathcal{E}_r^{n+r-1, q+2r-1}, \quad (5)$$

and the same formula holds for $\overline{\mathcal{E}}_r$.

Proof. In (2), set $n = -j$ and $q = i - j$. Then $i = q - n$ and $-i - j = 2n - q$. The differential in the original coordinates is $d_r : E_r^{i,j} \rightarrow E_r^{i+r, j-r+1}$, which becomes (5). $\square$

Lemma 2.4 (Three-row degeneration). *Suppose that the E_2 -page of either spectral sequence in Proposition 2.3 is supported in the rows $0 \leq q \leq 2$. Then every differential d_r with $r \geq 2$ vanishes.*

Proof. By (5), d_r raises q by $2r - 1 \geq 3$. Thus the source and target of neither an outgoing nor an incoming differential can both lie in the three rows $q = 0, 1, 2$. $\square$

We shall also use the following elementary convergence criterion.

Lemma 2.5 (Finite-diagonal convergence). *Let Y be equipped with a decreasing $\mathbf{Z}$ -indexed filtration $\{\mathrm{Fil}^n Y\}_{n \in \mathbf{Z}}$. Assume*

$$\mathrm{holim}_{n \rightarrow +\infty} \mathrm{Fil}^n Y \simeq 0, \quad \varinjlim_{n \rightarrow -\infty} \mathrm{Fil}^n Y \simeq Y.$$

If, for every integer t , the groups $\pi_t \mathrm{gr}^n Y$ vanish for all but finitely many n , then the associated exact-couple spectral sequence converges strongly to $\pi_ Y$. More precisely, the filtration*

$$F^n \pi_t Y := \mathrm{im}(\pi_t \mathrm{Fil}^n Y \rightarrow \pi_t Y)$$

is finite, exhaustive, and separated for every t .

Proof. Fix t . Choose integers $a < b$ such that

$$\pi_s \mathrm{gr}^n Y = 0 \quad \text{for } s \in \{t, t+1, t+2\} \text{ and } n \notin [a, b].$$

From the long exact sequence of

$$\mathrm{Fil}^{n+1} Y \longrightarrow \mathrm{Fil}^n Y \longrightarrow \mathrm{gr}^n Y$$

it follows that the transition maps on π_t and π_{t+1} are isomorphisms for $n < a$ and for $n > b$. At the negative end these groups are therefore constant; since homotopy groups commute with filtered colimits, exhaustiveness identifies the stable value of $\pi_t \mathrm{Fil}^n Y$ with $\pi_t Y$.

At the positive end the tower $\{\pi_{t+1} \mathrm{Fil}^n Y\}_{n \geq b}$ is constant, hence Mittag-Leffler. The Milnor exact sequence is

$$0 \longrightarrow \varprojlim_n \mathrm{olimits}_n^1 \pi_{t+1} \mathrm{Fil}^n Y \longrightarrow \pi_t \left(\mathrm{holim}_n \mathrm{Fil}^n Y \right) \longrightarrow \varprojlim_n \pi_t \mathrm{Fil}^n Y \longrightarrow 0.$$

The first term vanishes and completeness kills the middle term. Thus the stable positive-end value of $\pi_t \mathrm{Fil}^n Y$ is zero. Consequently $F^n \pi_t Y = \pi_t Y$ for $n \ll 0$ and $F^n \pi_t Y = 0$ for $n \gg 0$.

Only finitely many filtration quotients can therefore occur in total degree t . Restricting the exact couple to this finite range gives an ordinary finite spectral sequence whose abutment is the finite, exhaustive, separated filtration $F^\bullet \pi_t Y$. In particular there is no $\varprojlim^1$ or whole-plane obstruction, so the original spectral sequence converges strongly in degree t ; compare [4, Section 5]. Since t was arbitrary, the assertion follows. $\square$

3. THE SYNTOMIC INPUT

This section records the arithmetic input. Its proof is contained in the companion note [8]; repeating that proof here would obscure the part of the argument that is specific to TC. We retain the canonical maps and the multiplicative compatibility because they are needed to resolve the hidden extension at the prime two.

Retain the notation $C_n = \mathbf{Z}_p(n)(\mathcal{O}_K)$ and $\overline{C}_n = C_n \otimes_{\mathbf{Z}_p}^{\mathbf{L}} \mathbf{F}_p$. For $a \neq 0$, recall

$$W_a(K) = H^0(K, \mathbf{Q}_p/\mathbf{Z}_p(a)), \quad W_a(K)^\vee = \mathrm{Hom}_{\mathrm{cont}}(W_a(K), \mathbf{Q}_p/\mathbf{Z}_p).$$

Each $W_a(K)$ is a finite cyclic p -group, and

$$W_a(K)[p] \cong H^0(K, \mathbf{F}_p(a)). \quad (6)$$

Theorem 3.1 (Syntomic input). *The cohomology of C_n is given by*

n	$H^0(C_n)$	$H^1(C_n)$	$H^2(C_n)$
$n < 0$	0	0	0
$n = 0$	$\mathbf{Z}_p$	$\mathbf{Z}_p$	0
$n = 1$	0	$\widehat{\mathcal{O}_{K,p}^\times}$	0
$n \geq 2$	0	$H^1(K, \mathbf{Z}_p(n))$	$W_{1-n}(K)^\vee$.

(7)

There is no cohomology in any other degree. For $n \geq 2$,

$$H^1(K, \mathbf{Z}_p(n)) \cong \mathbf{Z}_p^N \oplus W_n(K)$$

noncanonically, while

$$\widehat{\mathcal{O}_{K,p}^\times} \cong \mathbf{Z}_p^N \oplus \mu_{p^\infty}(K)$$

noncanonically.

The identifications have the following canonical and multiplicative refinement.

(i) The generic restriction in weight one is the Kummer inclusion

$$H^1(C_1) = \widehat{\mathcal{O}_{K,p}^\times} \longrightarrow H^1(K, \mathbf{Z}_p(1)) = \widehat{K^\times}_p.$$

For $n \geq 2$ there is a canonical equivalence

$$C_n \simeq R\Gamma(K, \mathbf{Z}_p(n)).$$

(ii) For $a, b \geq 1$, the product on the graded syntomic algebra and the Galois cup product fit into a commutative diagram

$$\begin{array}{ccc} H^1(C_a) \otimes_{\mathbf{Z}_p} H^1(C_b) & \xrightarrow{\quad \smile \quad} & H^2(C_{a+b}) \\ \downarrow & & \downarrow \sim \\ H^1(K, \mathbf{Z}_p(a)) \otimes_{\mathbf{Z}_p} H^1(K, \mathbf{Z}_p(b)) & \xrightarrow{\quad \smile \quad} & H^2(K, \mathbf{Z}_p(a+b)). \end{array} \quad (8)$$

Here a vertical map in weight one is the Kummer inclusion above, and a vertical map in weight at least two is the canonical Galois comparison.

Proof. The cohomology table and the descriptions of the torsion groups are [8, Theorem 1.2 and Corollary 1.3]. The canonical comparison with the Geisser–Sato–Schneider twists and the localization triangle on $\mathrm{Spec}(\mathcal{O}_K)$ are [8, Proposition 1.6]. That comparison is induced by the canonical multiplicative finite-level equivalence of Bhatt–Mathew [2, Theorem 5.8]. The equivalences are compatible with reduction in the coefficient exponent, and taking the inverse limit in the ∞ -category of graded E_∞ -algebras preserves their multiplicative structure. Consequently the integral comparison carries the graded syntomic product to the usual product of Tate twists on the generic point, which gives diagram (8). Formula (6) and the cyclicity of $W_a(K)$ are proved in [8, Sections 4–5]. $\square$

Remark 3.2. The content suppressed here is precisely the material already established in [8]: the comparison with Sato twists, the closed-point purity calculation, the weight-one valuation and Brauer residues, and local Tate duality. None of those arguments is repeated below. The rest of the paper concerns the filtered passage from (7) to the homotopy groups of TC.

3.1. Derived reduction modulo p . The coefficient triangle

$$C_n \xrightarrow{p} C_n \longrightarrow \overline{C}_n \xrightarrow{+1}$$

gives functorial short exact sequences

$$0 \longrightarrow H^q(C_n)/p \longrightarrow H^q(\overline{C}_n) \longrightarrow H^{q+1}(C_n)[p] \longrightarrow 0. \quad (9)$$

For later use, note two canonical edge cases of (9). Since $H^0(C_n) = 0$ for $n \geq 1$, the Bockstein gives an isomorphism

$$\delta_n : H^0(\overline{C}_n) \xrightarrow{\sim} H^1(C_n)[p] \quad (n \geq 1). \quad (10)$$

Since $H^3(C_n) = 0$, reduction gives an isomorphism

$$H^2(C_n)/p \xrightarrow{\sim} H^2(\overline{C}_n) \quad (n \geq 0). \quad (11)$$

Lemma 3.3 (Cyclotomic period modulo p). *If p is odd and $d = [K(\mu_p) : K]$, then*

$$\varepsilon_a := \dim_{\mathbf{F}_p} H^0(K, \mathbf{F}_p(a)) = \begin{cases} 1, & d \mid a, \\ 0, & d \nmid a. \end{cases}$$

If $p = 2$, then $\varepsilon_a = 1$ for every $a \in \mathbf{Z}$.

Proof. The fixed field of the kernel of the mod- p cyclotomic character is $K(\mu_p)$, so its image has order d . The one-dimensional representation $\mathbf{F}_p(a)$ is trivial exactly when the a th power of this character is trivial. For $p = 2$, the target $\mathbf{F}_2^\times$ is trivial. $\square$

Proposition 3.4 (The complete finite-coefficient page). *The groups $H^q(\overline{C}_n)$ vanish for $q \notin \{0, 1, 2\}$. Their dimensions are*

n	$\dim H^0(\overline{C}_n)$	$\dim H^1(\overline{C}_n)$	$\dim H^2(\overline{C}_n)$
0	1	1	0
1	ε_1	$N + \varepsilon_1$	0
$n \geq 2$	ε_n	$N + \varepsilon_n + \varepsilon_{1-n}$	ε_{1-n} .

(12)

For $n \geq 2$ there is, canonically,

$$\overline{C}_n \simeq R\Gamma(K, \mathbf{F}_p(n)).$$

Proof. We derive every entry from the universal-coefficient sequence (9); no additional local-duality calculation is needed here.

For $n = 0$, the two nonzero integral groups are $H^0(C_0) = H^1(C_0) = \mathbf{Z}_p$. Since they are torsion-free, (9) gives

$$H^0(\overline{C}_0) \cong \mathbf{F}_p, \quad H^1(\overline{C}_0) \cong \mathbf{F}_p, \quad H^2(\overline{C}_0) = 0.$$

For $n = 1$, write noncanonically

$$H^1(C_1) \cong \mathbf{Z}_p^N \oplus \mu_{p^\infty}(K).$$

The finite cyclic group $\mu_{p^\infty}(K)$ has nonzero p -torsion exactly when $\mu_p \subset K$, equivalently when $\varepsilon_1 = 1$. Hence

$$\dim H^0(\overline{C}_1) = \dim H^1(C_1)[p] = \varepsilon_1,$$

while

$$\dim H^1(\overline{C}_1) = \dim H^1(C_1)/p = N + \varepsilon_1;$$

the degree-2 group vanishes.

Now let $n \geq 2$. By [Theorem 3.1](#),

$$H^1(C_n) \cong \mathbf{Z}_p^N \oplus W_n(K), \quad H^2(C_n) \cong W_{1-n}(K)^\vee$$

noncanonically, and both finite summands are cyclic. Using [\(6\)](#) and the fact that, for a finite cyclic p -group A , the vector spaces $A[p]$, A/p , and $A^\vee/p$ all have dimension one precisely when $A \neq 0$, we obtain

$$\dim H^1(C_n)[p] = \varepsilon_n, \quad \dim H^1(C_n)/p = N + \varepsilon_n,$$

and

$$\dim H^2(C_n)[p] = \dim H^2(C_n)/p = \varepsilon_{1-n}.$$

Substitution into [\(9\)](#) gives the third row of [\(12\)](#).

Finally, the canonical equivalence $C_n \simeq R\Gamma(K, \mathbf{Z}_p(n))$ for $n \geq 2$ identifies the cone of multiplication by p on C_n with the image under $R\Gamma(K, -)$ of

$$\mathbf{Z}_p(n) \xrightarrow{p} \mathbf{Z}_p(n) \longrightarrow \mathbf{F}_p(n).$$

Therefore

$$\overline{C}_n \simeq R\Gamma(K, \mathbf{F}_p(n))$$

canonically, as asserted. $\square$

4. FROM THE SYNTOMIC PAGE TO THE ABUTMENT

We now carry out the filtered-homotopy part of the argument. For either $Y = X_K$ or $Y = X_K/p$, let

$$F^n \pi_t Y := \operatorname{im}(\pi_t \operatorname{Fil}_{\operatorname{mot}}^n Y \longrightarrow \pi_t Y). \quad (13)$$

Once strong convergence is known, the associated graded of this filtration is

$$F^n \pi_t Y / F^{n+1} \pi_t Y \cong E_\infty^{n, 2n-t}(Y). \quad (14)$$

Theorem 4.1 (Degeneration and strong convergence). *The integral and mod- p motivic spectral sequences*

$$\mathcal{E}_2^{n,q} = H^q(C_n) \implies \pi_{2n-q} X_K, \quad \overline{\mathcal{E}}_2^{n,q} = H^q(\overline{C}_n) \implies \pi_{2n-q}(X_K/p)$$

converge strongly and degenerate at E_2 .

Proof. By [Theorem 3.1](#) and [Proposition 3.4](#), both pages are supported in the three rows $q = 0, 1, 2$. In the coordinates of [Proposition 2.3](#),

$$d_r : E_r^{n,q} \longrightarrow E_r^{n+r-1,q+2r-1}.$$

For $r \geq 2$ the cohomological degree rises by at least three, so every differential has either zero source or zero target. Hence $E_2 = E_\infty$.

For a fixed total degree t , a possible contribution satisfies

$$t = 2n - q, \quad q \in \{0, 1, 2\}.$$

There are at most two such pairs (n, q) : an odd total degree meets only the row $q = 1$, while an even total degree can meet the rows $q = 0$ and $q = 2$. Thus only finitely many graded pieces contribute to π_t . The BMS filtration is complete and exhaustive, and the same is true after smashing with the finite spectrum $\mathbb{S}/p$. The finite-diagonal criterion of [Lemma 2.5](#) therefore gives strong convergence and identifies the abutment filtration by [\(14\)](#). $\square$

Corollary 4.2 (Filtration quotients in each stem). *The following statements hold.*

- (i) *Integrally, every stem has at most one nonzero filtration quotient. There are canonical edge isomorphisms*

$$\pi_{-1}X_K \xrightarrow{\sim} H^1(C_0), \quad \pi_0X_K \xrightarrow{\sim} H^0(C_0), \quad (15)$$

$$\pi_{2m-1}X_K \xrightarrow{\sim} H^1(C_m), \quad \pi_{2m}X_K \xrightarrow{\sim} H^2(C_{m+1}) \quad (m \geq 1). \quad (16)$$

All stems below -1 vanish.

- (ii) *Modulo p , every nonzero odd stem has one filtration quotient. In a positive even stem $2m$, $m \geq 1$, the only possibly nontrivial filtration steps are*

$$F^m\pi_{2m}(X_K/p) = \pi_{2m}(X_K/p), \quad F^{m+2}\pi_{2m}(X_K/p) = 0,$$

and there is a canonical short exact sequence

$$0 \longrightarrow H^2(\overline{C}_{m+1}) \longrightarrow \pi_{2m}(X_K/p) \longrightarrow H^0(\overline{C}_m) \longrightarrow 0. \quad (17)$$

Both outer groups have dimension ε_m over $\mathbf{F}_p$.

Proof. Solve $t = 2n - q$ with $q \in \{0, 1, 2\}$. In the integral page, the row-0 terms vanish in positive weight, the row-2 term vanishes in weights 0 and 1, and the table [\(7\)](#) gives precisely the positions in [\(15\)](#)–[\(16\)](#). Since there is only one nonzero quotient, the corresponding edge map is an isomorphism.

In the mod- p page, an odd total degree meets only the row $q = 1$. The even total degree $2m$ meets the page at

$$(n, q) = (m, 0), \quad (n, q) = (m+1, 2).$$

Equation [\(14\)](#) gives [\(17\)](#). Finally,

$$\dim H^0(\overline{C}_m) = \varepsilon_m, \quad \dim H^2(\overline{C}_{m+1}) = \varepsilon_{-m} = \varepsilon_m.$$

$\square$

Proof of Theorem 1.2. Substitute Theorem 3.1 into the canonical edge isomorphisms (15)–(16). There is no additive extension because each integral stem has only one filtration quotient. $\square$

The next proposition is the key bridge between the spectral sequence and the stable homotopy of the Moore spectrum. It shows that the hidden extension in (17) is not merely abstract: it is exactly the coefficient extension attached to the cofiber sequence defining $\mathbb{S}/p$.

Proposition 4.3 (Agreement of the motivic and Moore extensions). *Let $m \geq 1$. The coefficient cofiber sequence*

$$X_K \xrightarrow{p} X_K \xrightarrow{i} X_K/p \xrightarrow{\partial} \Sigma X_K$$

gives a short exact sequence

$$0 \longrightarrow (\pi_{2m} X_K)/p \xrightarrow{i_*} \pi_{2m}(X_K/p) \xrightarrow{\partial} \pi_{2m-1} X_K[p] \longrightarrow 0. \quad (18)$$

Under the integral edge isomorphisms and the universal-coefficient maps (10)–(11), this is canonically the motivic extension (17). More precisely, the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\pi_{2m} X_K)/p & \xrightarrow{i_*} & \pi_{2m}(X_K/p) & \xrightarrow{\partial} & \pi_{2m-1} X_K[p] \longrightarrow 0 \\ & & \sim \downarrow & & \parallel & & \downarrow \sim \\ 0 & \longrightarrow & H^2(\overline{C}_{m+1}) & \longrightarrow & \pi_{2m}(X_K/p) & \longrightarrow & H^0(\overline{C}_m) \longrightarrow 0 \end{array} \quad (19)$$

commutes. The left vertical map is

$$(\pi_{2m} X_K)/p \xrightarrow[\text{(16)}]{\sim} H^2(C_{m+1})/p \xrightarrow[\text{(11)}]{\sim} H^2(\overline{C}_{m+1}),$$

and the right vertical map is

$$\pi_{2m-1} X_K[p] \xrightarrow[\text{(16)}]{\sim} H^1(C_m)[p] \xrightarrow{\delta_m^{-1}} H^0(\overline{C}_m),$$

where δ_m is the Bockstein isomorphism in (10).

Proof. By definition,

$$\text{Fil}^n(X_K/p) = \text{Fil}^n X_K \wedge \mathbb{S}/p.$$

Smashing the defining cofiber sequence of $\mathbb{S}/p$ with $\text{Fil}^n X_K$ gives, level by level, a cofiber sequence

$$\text{Fil}^n X_K \xrightarrow{p} \text{Fil}^n X_K \longrightarrow \text{Fil}^n(X_K/p) \longrightarrow \Sigma \text{Fil}^n X_K.$$

These sequences commute with the transition maps in n , so they form a cofiber sequence of filtered spectra. Passing to associated graded objects gives

$$C_n[2n] \xrightarrow{p} C_n[2n] \longrightarrow \overline{C}_n[2n] \longrightarrow C_n[2n+1].$$

Thus the induced morphism of exact couples acts on the E_2 -page by ordinary reduction in cohomological degree 2 and by the coefficient Bockstein from degree 0 to degree 1. We fix the boundary convention determined by the

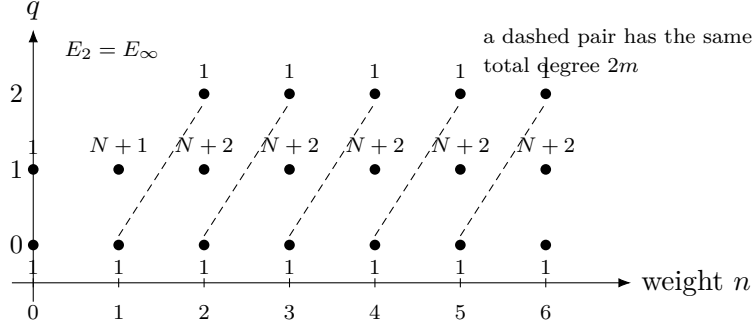


FIGURE 2. The complete mod-2 motivic page. The two points on the dashed diagonal through weights m and $m+1$ form the extension (17) in stem $2m$.

displayed cofiber sequence, so that the two vertical identifications in (19) make the square commute exactly. Reversing the convention changes both boundary maps by the same sign and leaves the extension class unchanged.

In total degree $2m$, the integral abutment has its unique nonzero filtration quotient in filtration $m+1$, whereas the mod- p abutment has possible quotients in filtrations m and $m+1$. Representing an integral class by $\text{Fil}^{m+1} X_K$ shows that $\text{im}(i_*) \subseteq F^{m+1}$. On the quotient

$$F^{m+1}/F^{m+2} = H^2(\overline{C}_{m+1}),$$

the map is exactly

$$H^2(C_{m+1})/p \xrightarrow{\sim} H^2(\overline{C}_{m+1}),$$

so it is an isomorphism. Hence $\text{im}(i_*) = F^{m+1}$.

Likewise, the boundary on $F^m/F^{m+1} = H^0(\overline{C}_m)$ is the Bockstein

$$H^0(\overline{C}_m) \xrightarrow{\sim} H^1(C_m)[p]$$

followed by the integral edge isomorphism. This proves the asserted commutative diagram without requiring any additional functoriality of the BMS construction in a coefficient spectrum. $\square$

For reference, the mod-2 page and the two terms lying on each positive even diagonal are displayed in Figure 2. The label at a point is its $\mathbf{F}_2$ -dimension.

5. ODD PRIMES

Lemma 5.1 (Odd-primary Moore exponent). *If p is odd, then*

$$p \text{ id}_{\mathbb{S}/p} = 0$$

in the stable homotopy category. Hence every group $\pi_t(Y \wedge \mathbb{S}/p)$ is annihilated by p .

Proof. Let

$$\mathbb{S} \xrightarrow{p} \mathbb{S} \xrightarrow{i} \mathbb{S}/p \xrightarrow{q} \Sigma\mathbb{S}$$

be the defining cofiber sequence. Since $(p \operatorname{id}_{\mathbb{S}/p}) \circ i = i \circ p = 0$, exactness after applying $[-, \mathbb{S}/p]$ shows that

$$p \operatorname{id}_{\mathbb{S}/p} = a \circ q$$

for some $a : \Sigma\mathbb{S} \rightarrow \mathbb{S}/p$. But a lies in $\pi_1(\mathbb{S}/p)$, and the long exact homotopy sequence identifies this group with zero: multiplication by odd p is an isomorphism on $\pi_1\mathbb{S} \cong \mathbf{Z}/2$ and is injective on $\pi_0\mathbb{S} \cong \mathbf{Z}$. Thus $a = 0$. Smashing this nullhomotopy with Y proves the final assertion. $\square$

Proposition 5.2 (Odd-prime extensions). *Let p be odd and $m \geq 1$. If $d \mid m$, then*

$$\pi_{2m}(X_K/p) \cong \mathbf{F}_p \oplus \mathbf{F}_p;$$

if $d \nmid m$, this group is zero.

Proof. By [Corollary 4.2](#), when $d \mid m$ the group has order p^2 and a filtration with two quotients $\mathbf{F}_p$. By [Lemma 5.1](#), the whole group has exponent p , so it cannot be cyclic of order p^2 . Hence it is $\mathbf{F}_p^2$. If $d \nmid m$, both filtration quotients vanish. $\square$

Proof of Theorem 1.1 for odd p . The groups in degrees -1 and 0 , the vanishing below -1 , and all odd stems are read directly from [Proposition 3.4](#) and [Corollary 4.2](#). The even stems are given by [Proposition 5.2](#). $\square$

Remark 5.3 (Periodic notation). For comparison with the usual generator notation, choose a nonzero class in $H^0(K, \mathbf{F}_p(d))$ and denote the corresponding formal period by β , of degree $2d$. Purely as periodic bookkeeping, and noncanonically as a graded $\mathbf{F}_p$ -vector space,

$$\pi_*(X_K/p) \cong \mathbf{F}_p[\beta]\{1, \lambda, \gamma, \lambda\gamma\} \oplus \bigoplus_{j=1}^d \bigoplus_{a=1}^e \bigoplus_{\ell=1}^f \mathbf{F}_p[\beta]\{\alpha_{a,\ell}^{(j)}\},$$

where

$$|\lambda| = -1, \quad |\gamma| = 2d + 1, \quad |\lambda\gamma| = 2d, \quad |\alpha_{a,\ell}^{(j)}| = 2j - 1.$$

The symbol $\mathbf{F}_p[\beta]$ records chosen periodic identifications of the mod- p Tate twists; it does not assert that X_K/p is a ring spectrum or that an element β acts on its homotopy groups. On the associated graded, $\lambda\gamma$ may be chosen as the cup product of the unramified class with the Kummer class of a uniformizer.

6. THE PRIME TWO

We now assume $p = 2$. The spectrum $\mathbb{S}/2$ is not used as a unital ring spectrum. All products are formed in the integral E_∞ -ring

$$X_K = \operatorname{TC}(\mathcal{O}_K; \mathbf{Z}_2),$$

and the Moore spectrum enters only through its cofiber sequence and its stable endomorphism relation.

6.1. The two-step extension and the Moore relation. Specializing [Proposition 4.3](#) to $p = 2$, for each $m \geq 1$ we have the canonical short exact sequence

$$0 \longrightarrow (\pi_{2m}X_K)/2 \xrightarrow{i_*} \pi_{2m}(X_K/2) \xrightarrow{\partial} \pi_{2m-1}X_K[2] \longrightarrow 0. \quad (20)$$

Both outer groups are one-dimensional over $\mathbf{F}_2$. Indeed,

$$\pi_{2m}X_K \cong W_{-m}(K)^\vee, \quad \text{tors } \pi_{2m-1}X_K \cong W_m(K),$$

and every $W_a(K)$, $a \neq 0$, is cyclic with nonzero two-torsion because the mod-2 cyclotomic character is trivial. Thus the middle group of (20) is either $\mathbf{F}_2^2$ or $\mathbf{Z}/4$.

Let

$$t_m \in \pi_{2m-1}X_K[2]$$

be the unique nonzero element, and choose $x_m \in \pi_{2m}(X_K/2)$ with $\partial x_m = t_m$. The extension is cyclic exactly when $2x_m \neq 0$.

Lemma 6.1 (The stable Moore relation). *For the cofiber sequence*

$$\mathbb{S} \xrightarrow{2} \mathbb{S} \xrightarrow{i} \mathbb{S}/2 \xrightarrow{q} \Sigma\mathbb{S}, \quad (21)$$

there is an identity in the stable homotopy category

$$2 \text{id}_{\mathbb{S}/2} = i\eta q, \quad (22)$$

where $\eta \in \pi_1\mathbb{S}$ is the stable Hopf map. Consequently, for every spectrum Y and every $x \in \pi_t(Y/2)$,

$$2x = i_*(\eta \partial x). \quad (23)$$

Proof. Write $M = \mathbb{S}/2$. Since $(2 \text{id}_M) \circ i = i \circ 2 = 0$, exactness after applying $[-, M]$ to (21) gives

$$2 \text{id}_M = a \circ q \quad \text{for some } a \in [\Sigma\mathbb{S}, M] = \pi_1 M.$$

The homotopy long exact sequence of (21) identifies $\pi_1 M \cong \mathbf{F}_2$, generated by $i\eta$. It remains only to rule out $2 \text{id}_M = 0$.

Smashing (21) with M identifies $M \wedge M$ with the cofiber of $2 \text{id}_M : M \rightarrow M$. If this map were null, then

$$M \wedge M \simeq M \vee \Sigma M.$$

Let $u \in \tilde{H}^0(M; \mathbf{F}_2)$ be the bottom class. We first verify that $\text{Sq}^1 u \neq 0$. For the coefficient sequence

$$0 \longrightarrow \mathbf{F}_2 \longrightarrow \mathbf{Z}/4 \longrightarrow \mathbf{F}_2 \longrightarrow 0,$$

the beginning of the cohomology long exact sequence is

$$\tilde{H}^0(M; \mathbf{Z}/4) \longrightarrow \tilde{H}^0(M; \mathbf{F}_2) \xrightarrow{\beta} \tilde{H}^1(M; \mathbf{F}_2).$$

The cellular cofiber description of M gives

$$\tilde{H}^0(M; \mathbf{Z}/4) = \ker(2 : \mathbf{Z}/4 \rightarrow \mathbf{Z}/4) = 2\mathbf{Z}/4,$$

and reduction modulo 2 sends this subgroup to zero. Hence the nonzero class u does not lift, so $\beta(u) \neq 0$. The coefficient Bockstein β is the stable operation Sq^1 ; put

$$v := \text{Sq}^1 u \neq 0 \in \tilde{H}^1(M; \mathbf{F}_2).$$

By the Künneth isomorphism and the Cartan formula,

$$\text{Sq}^2(u \otimes u) = \text{Sq}^1 u \otimes \text{Sq}^1 u = v \otimes v \neq 0 \quad \text{in } \tilde{H}^2(M \wedge M; \mathbf{F}_2).$$

On $M \vee \Sigma M$, however, the unique degree-zero class is $(u, 0)$. Steenrod operations act componentwise on the cohomology of a finite wedge, so its Sq^2 is $(\text{Sq}^2 u, 0) = (0, 0)$ because $\tilde{H}^2(M; \mathbf{F}_2) = 0$. This contradicts naturality of Steenrod operations under a stable equivalence. Hence 2id_M is nonzero. Therefore $2 \text{id}_M = i\eta q$.

Smashing the relation with Y and applying it to x gives

$$2x = i_*(\eta \partial x).$$

The possible sign is immaterial at the prime two. $\square$

Applying (23) to $Y = X_K$ and $x = x_m$ gives

$$2x_m = i_*(\eta t_m). \quad (24)$$

Since the left term of (20) is $(\pi_{2m} X_K)/2$, the extension problem is therefore reduced to computing the reduction modulo 2 of the integral product $\eta t_m \in \pi_{2m} X_K$.

6.2. The image of the Hopf class. We first identify the image of η in the canonical weight-one edge group.

Lemma 6.2 (The unit in algebraic K -theory). *Under the unit map*

$$\mathbb{S} \longrightarrow K(\mathcal{O}_K),$$

the class $\eta \in \pi_1 \mathbb{S} \cong \mathbf{Z}/2$ maps to $-1 \in K_1(\mathcal{O}_K) = \mathcal{O}_K^\times$.

Proof. The Barratt–Priddy theorem identifies the group completion of the symmetric monoidal groupoid of finite sets with $\Omega^\infty \mathbb{S}$; see [1]. The nontrivial class in $\pi_1 \mathbb{S}$ is represented by a transposition of a two-element set. The unit functor from finite sets to finitely generated projective $\mathcal{O}_K$ -modules sends this transposition to the corresponding permutation matrix. Its determinant is -1 , and the determinant identifies the class of this matrix in $K_1(\mathcal{O}_K)$ with the unit -1 . $\square$

Proposition 6.3 (The Hopf class is the unit -1). *Under the canonical edge isomorphism*

$$\pi_1 X_K \xrightarrow{\sim} H^1(C_1) \xrightarrow{\sim} \widehat{\mathcal{O}_K^\times},$$

the image of η under the unit $\mathbb{S} \rightarrow X_K$ is the class of -1 .

Proof. The cyclotomic trace is unital, so the image of η in X_K is the trace of the element -1 from [Lemma 6.2](#). We first show that this trace is nonzero.

The ring $\mathcal{O}_K$ is henselian along (2). The scheme $\mathrm{Spec}(\mathcal{O}_K/2)$ has the single residue field k , which is finite and hence perfect. Thus the invariant

$$\sup_{x \in \mathrm{Spec}(\mathcal{O}_K/2)} \log_2[k(x) : k(x)^2]$$

in the asymptotic comparison theorem of Clausen–Mathew–Morrow is 0. Their theorem gives, for every $r \geq 1$, an equivalence in degrees at least $\max(0, 1) = 1$; in particular,

$$\pi_1(K(\mathcal{O}_K)/2^r) \xrightarrow{\sim} \pi_1(\mathrm{TC}(\mathcal{O}_K)/2^r); \quad (25)$$

see [5, Theorem B].

The coefficient sequence for $K(\mathcal{O}_K)$ and the equality $K_0(\mathcal{O}_K) = \mathbf{Z}$ give

$$\pi_1(K(\mathcal{O}_K)/2^r) \cong K_1(\mathcal{O}_K)/2^r \cong \mathcal{O}_K^\times/(\mathcal{O}_K^\times)^{2^r}.$$

By [Lemma 2.1](#), the completion map induces

$$\mathrm{TC}(\mathcal{O}_K)/2^r \xrightarrow{\sim} X_K/2^r.$$

Moreover $\pi_0 X_K = \mathbf{Z}_2$ has no 2^r -torsion. The coefficient exact sequence therefore gives

$$\pi_1(\mathrm{TC}(\mathcal{O}_K)/2^r) \cong \pi_1(X_K/2^r) \cong (\pi_1 X_K)/2^r.$$

Choose r so large that K contains no primitive 2^{r+1} st root of unity. Then -1 is not a 2^r th power in K : if $u^{2^r} = -1$, then u would have exact order 2^{r+1} . Thus the class of -1 is nonzero on the left side of (25), and its trace is nonzero in $\pi_1 X_K$.

The element has order dividing two because it is the image of η . By [Theorem 3.1](#),

$$\pi_1 X_K \cong \widehat{\mathcal{O}_{K,2}^\times} \cong \mathbf{Z}_2^N \oplus \mu_{2^\infty}(K)$$

noncanonically. The first summand is torsion-free and the finite group of roots of unity in a local field is cyclic. Its unique element of order two is -1 . Therefore the nonzero order-two image of η is exactly -1 . $\square$

Under the Kummer inclusion from [Theorem 3.1](#), the element $-1 \in H^1(C_1)$ maps to the usual Kummer class in $H^1(K, \mathbf{Z}_2(1))$.

6.3. The cyclotomic Bockstein. For every $m \geq 1$, the coefficient sequence

$$0 \longrightarrow \mathbf{Z}_2(m) \xrightarrow{2} \mathbf{Z}_2(m) \longrightarrow \mathbf{F}_2(m) \longrightarrow 0 \quad (26)$$

has a connecting map

$$\delta_m : H^0(K, \mathbf{F}_2(m)) \longrightarrow H^1(K, \mathbf{Z}_2(m)).$$

The 2-adic cyclotomic character has infinite image, so $H^0(K, \mathbf{Z}_2(m)) = 0$ for $m \geq 1$; on the other hand the mod-2 cyclotomic character is trivial. We

identify every $\mathbf{F}_2(m)$ with $\mathbf{F}_2$ by its unique nonzero vector (equivalently by $(-1)^{\otimes m}$). Therefore the long exact sequence identifies

$$\delta_m : H^0(K, \mathbf{F}_2(m)) \xrightarrow{\sim} H^1(K, \mathbf{Z}_2(m))[2]. \quad (27)$$

For $m \geq 2$ this is the two-torsion in $H^1(C_m)$ via $C_m \simeq R\Gamma(K, \mathbf{Z}_2(m))$. For $m = 1$, the class $\delta_1(1)$ is the Kummer class of -1 , has valuation zero, and hence lies in the subgroup $H^1(C_1) = \widehat{\mathcal{O}_{K,2}^\times}$. Thus, in all weights,

$$t_m = \delta_m(1). \quad (28)$$

Lemma 6.4 (The mod-2 cyclotomic Bockstein). *Let*

$$\kappa(-1) \in H^1(K, \mathbf{F}_2(1))$$

be the Kummer class of -1 . Under the canonical identifications of all mod-2 Tate twists,

$$\bar{t}_m = (m \bmod 2) \kappa(-1) \quad \text{in } H^1(K, \mathbf{F}_2(m)). \quad (29)$$

Proof. Choose a basis e_m of the free rank-one module $\mathbf{Z}_2(m)$, so that $g(e_m) = \chi(g)^m e_m$. Its reduction $\bar{e}_m$ is G_K -fixed. The connecting class $\delta_m(e_m)$ from (26) is represented by the continuous cocycle

$$c_m(g) = \frac{\chi(g)^m - 1}{2} e_m. \quad (30)$$

Write $\chi(g) = 1 + 2a_g$ with $a_g \in \mathbf{Z}_2$. The binomial expansion modulo 4 gives

$$\chi(g)^m \equiv 1 + 2ma_g \pmod{4}.$$

Consequently the reduction of (30) modulo 2 is

$$g \mapsto (m \bmod 2)(a_g \bmod 2)\bar{e}_m.$$

It remains to identify the cocycle $g \mapsto a_g \bmod 2$. Choose a primitive fourth root of unity ζ_4 with $\zeta_4^2 = -1$. Since $g(\zeta_4) = \zeta_4^{\chi(g)}$,

$$\frac{g(\zeta_4)}{\zeta_4} = \zeta_4^{\chi(g)-1} = (-1)^{(\chi(g)-1)/2} = (-1)^{a_g}.$$

This is exactly the Kummer cocycle of -1 . Hence $a_g \bmod 2$ represents $\kappa(-1)$, proving (29). $\square$

6.4. Multiplicative edges and Galois cup products. The next step is the one for which the multiplicative structure of the motivic filtration is essential. We make the passage from filtered multiplication to the associated graded explicit.

Lemma 6.5 (Products through a one-step abutment filtration). *Let R be a ring spectrum with a decreasing multiplicative filtration, and suppose its spectral sequence converges strongly. Let $x \in F^a \pi_r R$ and $y \in F^b \pi_s R$. Then $xy \in F^{a+b} \pi_{r+s} R$, and the class of xy in*

$$F^{a+b} \pi_{r+s} R / F^{a+b+1} \pi_{r+s} R$$

is the product of the associated-graded classes of x and y . If $F^{a+b+1} \pi_{r+s} R = 0$, this associated-graded class determines xy itself.

Proof. Multiplicativity is a system of maps

$$\mathrm{Fil}^a R \otimes \mathrm{Fil}^b R \longrightarrow \mathrm{Fil}^{a+b} R$$

compatible with the maps to R . Passing to the quotients by $\mathrm{Fil}^{a+1} R$, $\mathrm{Fil}^{b+1} R$, and $\mathrm{Fil}^{a+b+1} R$ gives the product

$$\mathrm{gr}^a R \otimes \mathrm{gr}^b R \longrightarrow \mathrm{gr}^{a+b} R.$$

Taking homotopy groups and using strong convergence yields the claimed formula on the filtration of $\pi_* R$. If the next filtration step vanishes, the quotient map is an isomorphism. $\square$

Proposition 6.6 (The integral product on the edge). *For every $m \geq 1$, the canonical edge isomorphisms fit into a commutative diagram*

$$\begin{array}{ccc} \pi_1 X_K \otimes_{\mathbf{Z}_2} \pi_{2m-1} X_K & \xrightarrow{\quad \cdot \quad} & \pi_{2m} X_K \\ \sim \downarrow & & \downarrow \sim \\ H^1(C_1) \otimes_{\mathbf{Z}_2} H^1(C_m) & \xrightarrow{\quad \smile \quad} & H^2(C_{m+1}). \end{array} \quad (31)$$

After applying the generic comparison, the bottom product is the continuous Galois cup product

$$H^1(K, \mathbf{Z}_2(1)) \otimes H^1(K, \mathbf{Z}_2(m)) \longrightarrow H^2(K, \mathbf{Z}_2(m+1)).$$

Proof. The filtration positions are forced by [Corollary 4.2](#):

$$\begin{aligned} \pi_1 X_K &= F^1 \pi_1 X_K, & F^2 \pi_1 X_K &= 0, \\ \pi_{2m-1} X_K &= F^m \pi_{2m-1} X_K, & F^{m+1} \pi_{2m-1} X_K &= 0, \end{aligned}$$

and

$$\pi_{2m} X_K = F^{m+1} \pi_{2m} X_K, \quad F^{m+2} \pi_{2m} X_K = 0.$$

Apply [Lemma 6.5](#) with $(a, b) = (1, m)$. By the multiplicative refinement in [Theorem 2.2](#), the product on the associated graded is the graded syntomic cup product. This gives (31). The final assertion is exactly the multiplicative compatibility (8) in [Theorem 3.1](#). $\square$

Proposition 6.7 (The obstruction class). *Under the canonical isomorphisms*

$$(\pi_{2m} X_K)/2 \xrightarrow{\sim} H^2(C_{m+1})/2 \xrightarrow{\sim} H^2(K, \mathbf{F}_2(m+1)),$$

the reduction of ηt_m is

$$(m \bmod 2) \kappa(-1) \smile \kappa(-1). \quad (32)$$

Proof. By [Proposition 6.3](#), the lower-left edge class of η in (31) is the unit -1 , whose mod-2 generic restriction is $\kappa(-1)$. By [Lemma 6.4](#), the mod-2 reduction of the generic class of t_m is $(m \bmod 2) \kappa(-1)$. The product diagram (31) and the Galois compatibility in [Proposition 6.6](#) therefore give (32).

Finally, $m+1 \geq 2$, so [Theorem 3.1](#) gives the canonical equivalence

$$C_{m+1} \simeq R\Gamma(K, \mathbf{Z}_2(m+1))$$

and also gives $H^3(C_{m+1}) = 0$. Applying the coefficient triangle to this complex, or equivalently using (11), yields canonical isomorphisms

$$H^2(K, \mathbf{Z}_2(m+1))/2 \xrightarrow{\sim} H^2(\overline{C}_{m+1}) \xrightarrow{\sim} H^2(K, \mathbf{F}_2(m+1)).$$

This proves the second displayed identification without invoking a separate cohomological-dimension assertion for compact coefficients. $\square$

6.5. The Hilbert symbol and the extension. Because the mod-2 cyclotomic character is trivial, there is a unique isomorphism of one-dimensional G_K -modules $\mathbf{F}_2(a) \cong \mathbf{F}_2(b)$ for every pair of integers a, b . Under the resulting identification

$$H^2(K, \mathbf{F}_2(m+1)) \cong H^2(K, \mathbf{F}_2(2)),$$

the class in (32) is the usual cup square of the Kummer class of -1 .

Lemma 6.8 (The Hamilton quaternion class). *The cup square*

$$\kappa(-1) \smile \kappa(-1) \in H^2(K, \mathbf{F}_2(2))$$

is nonzero if and only if $[K : \mathbf{Q}_2]$ is odd.

Proof. Kummer theory identifies the cup product with the norm-residue symbol

$$\{-1, -1\}_K,$$

which is the Brauer class of the quaternion algebra $(-1, -1)_K$. By the cyclic-algebra norm criterion, $(-1, -1)_{\mathbf{Q}_2}$ splits precisely when -1 is a norm from $\mathbf{Q}_2(i)$, equivalently when $x^2 + y^2 = -1$ has a solution in $\mathbf{Q}_2$; see [7, Chapter XIV, §2]. We verify directly that no such solution exists. Suppose $x^2 + y^2 = -1$ with $x, y \in \mathbf{Q}_2$, and put $r = \min\{v_2(x), v_2(y)\}$. If $r < 0$, write $x = 2^r x_0$ and $y = 2^r y_0$ with $x_0, y_0 \in \mathbf{Z}_2$ and at least one of x_0, y_0 a unit. Then $x_0^2 + y_0^2 = -2^{-2r}$ is divisible by 4, whereas modulo 4 the left side is 1 or 2, a contradiction. Hence $r \geq 0$, so $x, y \in \mathbf{Z}_2$; reducing the original equation modulo 4 gives another contradiction, since a sum of two squares is 0, 1, or 2, never $-1 \equiv 3$. Thus $(-1, -1)_{\mathbf{Q}_2}$ is the nonzero element of $\text{Br}(\mathbf{Q}_2)[2]$ and has invariant $1/2 \in \mathbf{Q}/\mathbf{Z}$.

The algebra over K is obtained by restriction from $\mathbf{Q}_2$. The local invariant isomorphism and its compatibility with restriction give

$$\text{inv}_K(\text{res}_{K/\mathbf{Q}_2} \alpha) = [K : \mathbf{Q}_2] \text{inv}_{\mathbf{Q}_2}(\alpha).$$

Consequently

$$\text{inv}_K((-1, -1)_K) = [K : \mathbf{Q}_2] \cdot \frac{1}{2} \pmod{\mathbf{Z}},$$

which is nonzero exactly when the degree is odd. For the local invariant and the restriction formula, see [7, Chapters XIII–XIV]. $\square$

Theorem 6.9 (All hidden 2-extensions). *For $m \geq 1$, let x_m be any lift of t_m through (20). Then*

$$2x_m \neq 0 \iff m \text{ and } [K : \mathbf{Q}_2] \text{ are both odd.}$$

Consequently,

$$\pi_{2m}(X_K/2) \cong \begin{cases} \mathbf{Z}/4, & m \text{ odd and } [K : \mathbf{Q}_2] \text{ odd,} \\ \mathbf{F}_2 \oplus \mathbf{F}_2, & \text{otherwise.} \end{cases} \quad (33)$$

These are the only hidden additive extensions in the mod-2 motivic spectral sequence.

Proof. By (24),

$$2x_m = i_*(\eta t_m).$$

The injection i_* factors through $(\pi_{2m}X_K)/2$, so this element is nonzero exactly when the reduction of ηt_m modulo 2 is nonzero. By Proposition 6.7, that reduction is zero for even m , while for odd m it is the cup square $\kappa(-1)^2$. By Lemma 6.8, the latter is nonzero exactly when $[K : \mathbf{Q}_2]$ is odd.

The group in the middle of (20) has order four. If $2x_m \neq 0$, then x_m has order four and the group is cyclic. If $2x_m = 0$, every element is killed by two and the group is $\mathbf{F}_2^2$. All odd stems, as well as degrees -1 and 0 , have a single filtration quotient by Corollary 4.2; hence no further additive extension is possible. $\square$

Completion of the proof of Theorem 1.1. The dimensions of the E_∞ -terms are given by Proposition 3.4. Strong convergence and the location of the filtration quotients are given by Theorem 4.1 and Corollary 4.2. Odd stems and the two low stems therefore have the groups stated in Theorem 1.1; the positive even stems are given by Theorem 6.9. $\square$

Remark 6.10 (No product on the Moore reduction). The cup product in Propositions 6.6 and 6.7 is formed in the integral E_∞ -ring X_K . It is then mapped to $(\pi_{2m}X_K)/2$ and inserted into the extension by the stable relation $2\text{id}_{\mathbb{S}/2} = i\eta q$. No unital multiplication on $\mathbb{S}/2$ or on $X_K/2$ is assumed.

7. EXAMPLES

Example 7.1 (The base field $\mathbf{Q}_p$, p odd). Here $N = 1$ and $d = p - 1$. Thus, for $m \geq 1$,

$$\pi_{2m}\text{TC}(\mathbf{Z}_p; \mathbf{F}_p) \cong \begin{cases} \mathbf{F}_p^2, & p-1 \mid m, \\ 0, & p-1 \nmid m. \end{cases}$$

Moreover,

$$\pi_1 \cong \mathbf{F}_p, \quad \pi_{2m-1} \cong \mathbf{F}_p^{1 + \mathbf{1}_{p-1 \mid m} + \mathbf{1}_{p-1 \mid (1-m)}} \quad (m \geq 2).$$

Example 7.2 (The field $\mathbf{Q}_2$). Here $N = 1$. Hence

$$\pi_{4r+2}\text{TC}(\mathbf{Z}_2; \mathbf{F}_2) \cong \mathbf{Z}/4, \quad \pi_{4r+4}\text{TC}(\mathbf{Z}_2; \mathbf{F}_2) \cong \mathbf{F}_2^2 \quad (r \geq 0),$$

while $\pi_1 \cong \mathbf{F}_2^2$ and $\pi_{2m-1} \cong \mathbf{F}_2^3$ for $m \geq 2$.

Example 7.3 (Even-degree extensions of $\mathbf{Q}_2$). If $[K : \mathbf{Q}_2]$ is even, then $(-1, -1)_K$ is split. Thus every positive even group $\pi_{2m}(X_K/2)$ is $\mathbf{F}_2^2$, and no $\mathbf{Z}/4$ occurs.

REFERENCES

- [1] M. G. Barratt and S. Priddy, *On the homology of non-connected monoids and their associated groups*, Comment. Math. Helv. **47** (1972), 1–14.
- [2] B. Bhatt and A. Mathew, *Syntomic complexes and p -adic étale Tate twists*, Forum Math. Pi **11** (2023), e1.
- [3] B. Bhatt, M. Morrow, and P. Scholze, *Topological Hochschild homology and integral p -adic Hodge theory*, Publ. Math. Inst. Hautes Études Sci. **129** (2019), 199–310.
- [4] J. M. Boardman, *Conditionally convergent spectral sequences*, in Homotopy invariant algebraic structures (Baltimore, MD, 1998), Contemp. Math. **239**, Amer. Math. Soc., 1999, 49–84.
- [5] D. Clausen, A. Mathew, and M. Morrow, *K -theory and topological cyclic homology of henselian pairs*, J. Amer. Math. Soc. **34** (2021), 411–473.
- [6] R. Liu and G. Wang, *Topological cyclic homology of local fields*, Invent. Math. **230** (2022), 851–932.
- [7] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics 67, Springer, 1979.
- [8] W. Yang, *Syntomic Tate twists over p -adic local fields and the cyclotomic torsion groups*, preprint, version of June 2026, [available online](#).